

Project Development Phase

Functional & Performance Testing Benchmark Report

Attribute	Project Information
Project Title	Auto Ticket Classification using Flow Designer
Phase / Stage	5. Project Development Phase
Document Title	Functional & Performance Testing Report
Platform & Scope	ServiceNow Utah / Vancouver / Washington DC (Global Scope)
Team ID	SWTID-2026-4988

1. Testing Scope & Performance Objectives

The objective of the Development & Testing Phase is to empirically validate that the ServiceNow Flow Designer automation ("Auto Classify School IT Tickets") and the backend schema meet all functional requirements, security guardrails, and operational throughput benchmarks under both baseline and high-volume campus conditions.

2. Performance Benchmark Metrics & Latency Profiling

System latency tests were executed in the development instance to record execution timings across the end-to-end incident lifecycle:

Subsystem / Operation	Target SLA Threshold	Observed Benchmark	Performance Evaluation	Engineering Notes
Form Submission & Database Insert	< 500 ms	85 ms - 140 ms	EXCELLENT	Instantaneous database commit on table u_incident_workflow; auto-number generated cleanly.
Flow Designer Event Trigger Detection	< 1000 ms	180 ms - 260 ms	EXCELLENT	Workflow engine detects "Category is empty" condition asynchronously without user UI locking.
Keyword Parsing & Conditional Branching	< 300 ms	45 ms - 80 ms	EXCELLENT	String regex evaluation in memory executes instantaneously across all 4 conditional branches.
Record Field Update (Category & Subcategory)	< 500 ms	95 ms - 150 ms	EXCELLENT	GlideRecord update commits values and triggers dictionary validation successfully.

Subsystem / Operation	Target SLA Threshold	Observed Benchmark	Performance Evaluation	Engineering Notes
Outbound Email Payload Generation	< 500 ms	110 ms - 190 ms	EXCELLENT	sys_email record generated with dynamic Caller data pills and queued for SMTP delivery.
Total End-to-End Triage Latency	< 2800 ms	515 ms - 820 ms	SUPERIOR	Full automated triage completes in ~0.7 seconds, representing a 99.9% speedup over manual triage.

3. Concurrent Load & Stress Testing

Test Scenario	Concurrent Transactions	Success Rate	Observed Anomalies / Collisions
Simulated Morning Rush (Wi-Fi Surge)	25 simultaneous ticket submissions	100% (25/25)	Zero row-level database locks; all 25 records classified as Network / Wi-Fi within 1.2s.
Mixed Incident Load (All 4 Categories)	50 simultaneous ticket submissions	100% (50/50)	Zero flow execution timeouts; all 50 auto-numbers (INC005xx) assigned sequentially.
Unrecognized Keyword Fallback Load	15 simultaneous vague submissions	100% (15/15)	All 15 bypassed conditional updates cleanly without error, preserving empty Category for manual desk.

4. Resource Utilization & Platform Impact

Because Flow Designer is natively embedded in the ServiceNow platform runtime, it executes with minimal overhead:

- CPU & Node Memory Impact: Under 0.05% per execution.
- Database Transaction Overhead: Consumes only two atomic GlideRecord operations (one insert, one update).
- Storage Footprint: Extremely compact; no temporary bloat tables or external cache layers required.